Math for CS & AI: Homework 11

Yifan Chen

Problem 5.19

Let $L \geq 3$. An L-layer ReLU network is a neural network with L linear layers, where each intermediate layer is followed by a ReLU activation function $[x]_+ := \max\{x, 0\}$. When applied to a vector $\boldsymbol{x} = (x_1, \ldots, x_d) \in \mathbb{R}^d$, the ReLU activation function is applied element-wise:

$$[\boldsymbol{x}]_+ = (\max\{x_1, 0\}, \ldots, \max\{x_d, 0\}) .$$

More specifically, an L-layer ReLU network is parameterized by a sequence of weight matrices $(\boldsymbol{W}_1, \ldots, \boldsymbol{W}_L)$. Given an input $\boldsymbol{x} \in \mathbb{R}^d$, we write $\boldsymbol{x}^{(0)} := \boldsymbol{x}$, and then for each $\ell \in [L-1]$, the output of the ℓ-th layer is given by

$$\boldsymbol{x}^{(\ell)} := \left[\boldsymbol{W}_\ell \boldsymbol{x}^{(\ell-1)}\right]_+ .$$

Finally, the output of the L-th layer, which is also the output of the network, is given by

$$f(\boldsymbol{x}) := \boldsymbol{W}_L \boldsymbol{x}^{(L-1)}.$$

The first and last layers are called the input layer and output layer, respectively. The intermediate layers are called hidden layers. Let d and k be the dimensions of the input and output, respectively, and let m be the parameter that denotes the width of the network. The weight matrix in the input layer has shape $m \times d$, the weight matrices in the hidden layers have shape $m \times m$, and the weight matrix in the output layer has shape $k \times m$. We use d_ℓ to denote the dimension of $\boldsymbol{x}^{(\ell)}$, so that $d_0 = d$ and $d_\ell = m$ for $\ell \in [L-1]$.

Having introduced all this background, we finally ask the following question: how should we initialize a deep ReLU network? In deep learning, weight matrices are usually initialized randomly, and the network is then trained by gradient descent. A typical choice is to sample each entry of each weight matrix $\boldsymbol{W}_\ell$ independently from

$$\mathcal{N}\left(0, c \cdot d_{\ell-1}^{-\alpha}\right),$$

where $c > 0$ and $\alpha > 0$ are constants.

(a) Fix an input $\boldsymbol{x} \in \mathbb{R}^d$. For every $\ell \in [L-1]$, express $\mathbb{E}\left[\|\boldsymbol{x}^{(\ell)}\|_2^2\right]$ in terms of $\|\boldsymbol{x}\|_2^2$, k, d, m, c, α, and ℓ. Here the expectation is taken over the random draw of the weight matrices.

Hint: If the distribution of X is symmetric around 0, then $\mathbb{E}\left[[X]_+^2\right] = \frac{1}{2}\mathbb{E}\left[X^2\right]$.

(b) An ideal initialization should ensure that the outputs of intermediate layers are not too small and not too large. Find a pair (c, α) such that for every fixed input $\boldsymbol{x} \in \mathbb{R}^d \setminus \{0\}$ and every coordinate $i \in [m]$,

$$\mathbb{E}\left[\left(x_i^{(L-1)}\right)^2\right]$$

converges to a non-zero constant as $m \to \infty$ and $L \to \infty$, where $x_i^{(L-1)}$ is the i-th coordinate of $\boldsymbol{x}^{(L-1)}$.

Note: The choice of c and α is unique, and is usually referred to as the *He initialization* or *Kaiming initialization*. This initialization is named after Kaiming He [HZRS15].

Answer:

For part (a), we have

$$\begin{aligned}
x_i^{(\ell)} &= \left[\sum_{j=1}^{d_{\ell-1}} \boldsymbol{W}_{\ell,i,j} x_j^{(\ell-1)}\right]_+, \\
\mathbb{E}\left[\|\boldsymbol{x}^{(\ell)}\|_2^2\right] &= \mathbb{E}\left[\sum_{i=1}^{d_\ell} \left(x_i^{(\ell)}\right)^2\right] \\
&= \mathbb{E}\left[\sum_{i=1}^{d_\ell} \left[\sum_{j=1}^{d_{\ell-1}} \boldsymbol{W}_{\ell,i,j} x_j^{(\ell-1)}\right]_+^2\right] \\
&= \frac{1}{2}\mathbb{E}\left[\sum_{i=1}^{d_\ell} \left(\sum_{j=1}^{d_{\ell-1}} \boldsymbol{W}_{\ell,i,j} x_j^{(\ell-1)}\right)^2\right].
\end{aligned}$$

Since $\boldsymbol{W}_{\ell,i,j} \sim \mathcal{N}(0, c \cdot d_{\ell-1}^{-\alpha})$ independently, the linear combination inside the brackets is symmetric around zero, which justifies the step above. For the cross terms, we have

$$\mathbb{E}\left[\boldsymbol{W}_{\ell,i,j}\boldsymbol{W}_{\ell,i,k} x_j^{(\ell-1)} x_k^{(\ell-1)}\right] = \mathbb{E}\left[\boldsymbol{W}_{\ell,i,j}\right] \cdot \mathbb{E}\left[\boldsymbol{W}_{\ell,i,k}\right] \cdot \mathbb{E}\left[x_j^{(\ell-1)} x_k^{(\ell-1)}\right] = 0,$$

because $\boldsymbol{W}_{\ell,i,j}$ and $\boldsymbol{W}_{\ell,i,k}$ are independent zero-mean random variables, and both are independent of the previous layer output $\boldsymbol{x}^{(\ell-1)}$. Therefore, we obtain:

$$\begin{aligned}
\mathbb{E}\left[\|\boldsymbol{x}^{(\ell)}\|_2^2\right] &= \frac{1}{2}\sum_{i=1}^{d_\ell}\sum_{j=1}^{d_{\ell-1}} \mathbb{E}\left[\boldsymbol{W}_{\ell,i,j}^2 \left(x_j^{(\ell-1)}\right)^2\right] \\
&= \frac{1}{2}\sum_{i=1}^{d_\ell}\sum_{j=1}^{d_{\ell-1}} \mathbb{E}\left[\boldsymbol{W}_{\ell,i,j}^2\right] \mathbb{E}\left[\left(x_j^{(\ell-1)}\right)^2\right] \\
&= \frac{1}{2}\sum_{i=1}^{d_\ell} c \cdot d_{\ell-1}^{-\alpha} \cdot \mathbb{E}\left[\|\boldsymbol{x}^{(\ell-1)}\|_2^2\right] \\
&= \frac{c \cdot d_\ell}{2 \cdot d_{\ell-1}^{\alpha}} \mathbb{E}\left[\|\boldsymbol{x}^{(\ell-1)}\|_2^2\right].
\end{aligned}$$

Note that $\mathbb{E}\left[\boldsymbol{W}_{\ell,i,j}^2\right] = \mathrm{Var}\left(\boldsymbol{W}_{\ell,i,j}\right)$ because $\mathbb{E}\left[\boldsymbol{W}_{\ell,i,j}\right] = 0$. Therefore, for any $1 \le \ell < L$, we have:

$$\mathbb{E}\left[\|\boldsymbol{x}^{(\ell)}\|_2^2\right] = d^{-\alpha} m^{\ell(1-\alpha)+\alpha} \cdot \left(\frac{c}{2}\right)^{\ell} \cdot \|\boldsymbol{x}\|_2^2.$$

For part (b), by the above arguments, we have

$$\mathbb{E}\left[\left(x_i^{(L-1)}\right)^2\right] = d^{-\alpha} m^{(L-2)(1-\alpha)} \cdot \left(\frac{c}{2}\right)^{L-1} \cdot \|\boldsymbol{x}\|_2^2.$$

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To ensure it converges to some non-zero constant, we must set $\alpha = 1$ and $c = 2$, and hence

$$\mathbb{E}\left[\left(x_i^{(L-1)}\right)^2\right] = \frac{1}{d} \cdot \|\boldsymbol{x}\|_2^2.$$

Problem 6.1

Let $d \geq 2$ and fix a unit vector $\boldsymbol{u} \in \mathbb{R}^d$. Let $\mathcal{U}_{\mathbb{S}}$ be the uniform distribution on the unit sphere $\mathbb{S}^{d-1}$. Prove that if $\boldsymbol{x} \sim \mathcal{U}_{\mathbb{S}}$, then as $d \to \infty$,

$$\sqrt{d}\langle \boldsymbol{u}, \boldsymbol{x}\rangle \xrightarrow{d} \mathcal{N}(0,1).$$

Answer:

Without loss of generality, we assume that $\boldsymbol{u} = (1, 0, 0, \ldots, 0)^\top$ by the rotational symmetry of the unit sphere. Under this assumption, the inner product simplifies to $\langle \boldsymbol{u}, \boldsymbol{x}\rangle = x_1$. For a fixed value $t \in [-1, 1]$, we define the slice:

$$A_t := \{\boldsymbol{x} \in \mathbb{S}^{d-1} : x_1 = t\},$$

which is a $(d-2)$-dimensional sphere of radius $\sqrt{1-t^2}$. Then by Theorem 5.41,

$$\mu(A_t) = \left(\sqrt{1-t^2}\right)^{d-2} \cdot S_{d-2} = \left(1-t^2\right)^{(d-2)/2} \cdot S_{d-2}.$$

For every $a < b$, we have:

$$\Pr[\langle \boldsymbol{u}, \boldsymbol{x}\rangle \in (a,b)] = \frac{\int_a^b \mu(A_t) \frac{1}{\sqrt{1-t^2}}\,\mathrm{d}t}{\int_{-1}^1 \mu(A_t) \frac{1}{\sqrt{1-t^2}}\,\mathrm{d}t} = \frac{\int_a^b \left(1-t^2\right)^{(d-3)/2}\,\mathrm{d}t}{\int_{-1}^1 \left(1-t^2\right)^{(d-3)/2}\,\mathrm{d}t}.$$

Let

$$I := \int_{-1}^1 \left(1-t^2\right)^{(d-3)/2}\,\mathrm{d}t.$$

Let $Y_d = \sqrt{d}\langle \boldsymbol{u}, \boldsymbol{x}\rangle$. Then Y_d has the PDF:

$$f_{Y_d}(y) = \begin{cases} \frac{1}{I\sqrt{d}} \left(1 - \frac{y^2}{d}\right)^{(d-3)/2}, & \text{if } |y| \leq \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases}$$

As $d \to \infty$, for any fixed $y \in \mathbb{R}$, we have

$$\begin{aligned} \left(1 - \frac{y^2}{d}\right)^{(d-3)/2} &= \exp\left(\frac{d-3}{2} \cdot \ln\left(1 - \frac{y^2}{d}\right)\right) \\ &= \exp\left(\frac{d-3}{2} \cdot \left(-\frac{y^2}{d} + \mathcal{O}\left(\frac{1}{d^2}\right)\right)\right) \\ &\to e^{-y^2/2}. \end{aligned}$$

This limit matches the standard Gaussian density up to a normalizing constant. Since f_{Y_d} is a probability density function, $\int_{-\infty}^{\infty} f_{Y_d}(y)\,\mathrm{d}y = 1$ for all d. Applying the dominated convergence theorem to interchange

limit and integral, we obtain

$$\lim_{d\to\infty} \frac{1}{I\sqrt{d}} \int_{-\sqrt{d}}^{\sqrt{d}} \left(1 - \frac{y^2}{d}\right)^{(d-3)/2} \mathrm{d}y = \lim_{d\to\infty} \frac{1}{I\sqrt{d}} \cdot \sqrt{2\pi} = 1,$$

which implies

$$\lim_{d\to\infty} \frac{1}{I\sqrt{d}} = \frac{1}{\sqrt{2\pi}}.$$

Therefore, for every fixed $y \in \mathbb{R}$,

$$\lim_{d\to\infty} f_{Y_d}(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2},$$

which is exactly the PDF of $\mathcal{N}(0,1)$. By Theorem 6.10, convergence of PDFs implies convergence in distribution. Hence,

$$\sqrt{d}\langle \boldsymbol{u}, \boldsymbol{x}\rangle \xrightarrow{d} \mathcal{N}(0,1).$$

Problem 6.2

Let $\boldsymbol{A} \in \mathbb{R}^{n\times d}$ be a Gaussian random matrix, where every entry of $\boldsymbol{A}$ is sampled independently from $\mathcal{N}(0, 1/n)$.

(a) Fix a vector $\boldsymbol{x} \in \mathbb{R}^d$. Prove that as $n \to \infty$,

$$\boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{x} \xrightarrow{d} \boldsymbol{x}.$$

(b) Fix another vector $\boldsymbol{y} \in \mathbb{R}^d$. Which distribution does the following random variable converge in distribution to as $n \to \infty$?

$$\sqrt{n}\left(\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{A}\boldsymbol{y}\rangle - \langle \boldsymbol{x}, \boldsymbol{y}\rangle\right)$$

Answer:

For part (a), we proceed coordinate-wise. Let $\boldsymbol{B} = \sqrt{n}\boldsymbol{A}$; then each $B_{i,j} \sim \mathcal{N}(0,1)$ independently. For the i-th coordinate, we have

$$(\boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{x})_i = \sum_{j=1}^{n}\sum_{k=1}^{d} A_{j,i}A_{j,k}x_k = \sum_{k=1}^{d} x_k \cdot \frac{1}{n}\sum_{j=1}^{n} B_{j,i}B_{j,k}.$$

We have $\mathbb{E}[B_{j,i}B_{j,k}] = \mathbb{1}_{[i=k]}$ and $\mathrm{Var}(B_{j,i}B_{j,k}) < +\infty$. By the Weak Law of Large Numbers, this implies that

$$(\boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{x})_i \xrightarrow{d} x_i,$$

which directly implies that

$$\boldsymbol{A}^\top \boldsymbol{A}\boldsymbol{x} \xrightarrow{d} \boldsymbol{x}.$$

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For part (b), we expand the expression as

$$\begin{aligned}
&\sqrt{n}\left(\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{A}\boldsymbol{y}\rangle - \langle \boldsymbol{x}, \boldsymbol{y}\rangle\right) \\
&= \sqrt{n}\left(\sum_{i=1}^{n}\sum_{j=1}^{d} A_{i,j}x_j \sum_{k=1}^{d} A_{i,k}y_k - \sum_{j=1}^{d} x_j y_j\right) \\
&= \sqrt{n}\left(\sum_{j=1}^{d}\sum_{k=1}^{d} x_j y_k \cdot \frac{1}{n}\sum_{i=1}^{n} B_{i,j}B_{i,k} - \sum_{j=1}^{d} x_j y_j\right) \\
&= 2\sum_{j=1}^{d}\sum_{k=j+1}^{d} x_j y_k \cdot \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} B_{i,j}B_{i,k} - 0\right) + \sum_{j=1}^{d} x_j y_j \cdot \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} B_{i,j}^2 - 1\right).
\end{aligned}$$

For the cross terms with $j \neq k$, we have $\mathbb{E}[B_{i,j}B_{i,k}] = 0$ and $\mathrm{Var}(B_{i,j}B_{i,k}) = 1$. For the diagonal terms with $j = k$, we have $\mathbb{E}\left[B_{i,j}^2\right] = 1$. To compute the variance, we calculate the fourth moment

$$\begin{aligned}
\mathbb{E}\left[B_{i,j}^4\right] &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} x^4 e^{-x^2/2}\mathrm{d}x \\
&= \frac{1}{\sqrt{2\pi}}\left(-x^3 e^{-x^2/2}\Big|_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} 3x^2 e^{-x^2/2}\mathrm{d}x\right) \\
&= 3.
\end{aligned}$$

Therefore, $\mathrm{Var}\left(B_{i,j}^2\right) = 3 - 1 = 2$. By the Central Limit Theorem, we obtain

$$\sqrt{n}\left(\langle \boldsymbol{A}\boldsymbol{x}, \boldsymbol{A}\boldsymbol{y}\rangle - \langle \boldsymbol{x}, \boldsymbol{y}\rangle\right) \xrightarrow{d} \mathcal{N}\left(0, \|\boldsymbol{x}\|_2^2\|\boldsymbol{y}\|_2^2 + \langle \boldsymbol{x}, \boldsymbol{y}\rangle^2\right).$$

Problem 6.3

Let $X_1, \ldots, X_n$ be n i.i.d. random samples from $\mathcal{N}(0, \sigma^2)$, and let $X^{(n)}$ be the maximum of $X_1, \ldots, X_n$. Find two sequences $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$ such that

$$\frac{X^{(n)} - a_n}{b_n} \xrightarrow{d} \text{Gumbel}(0, 1).$$

Here $\text{Gumbel}(0, 1)$ is the standard Gumbel distribution, whose PDF and CDF are given by

$$f_G(x) = e^{-\left(x+e^{-x}\right)} \quad \text{and} \quad F_G(x) = e^{-e^{-x}}.$$

Note: You may meet this Gumbel distribution once more in the Gumbel-softmax trick, a useful technique in machine learning for backpropagating through discrete sampling.

Answer:

The convergence requires that

$$\Pr\left[\frac{X^{(n)} - a_n}{b_n} \leq x\right] = \left[F_X(a_n + b_n x)\right]^n \to F_G(x),$$

where F_X is the CDF of $\mathcal{N}(0, \sigma^2)$. Let $z_n = \frac{a_n + b_n x}{\sigma}$. Taking the logarithm of both sides, the left-hand side satisfies

$$n \log F_X(a_n + b_n x) = n \log\left[1 - \bar{\Phi}\left(z_n\right)\right] \to -n\bar{\Phi}\left(z_n\right) \to -\frac{n}{\sqrt{2\pi}} \cdot \frac{e^{-z_n^2/2}}{z_n},$$

where we assume that $z_n \to +\infty$. Taking the logarithm again, we obtain

$$z_n^2 = 2\log n - 2\log z_n - \log(2\pi) + 2x + o(1).$$

Substituting $z_n \sim \sqrt{2\log n}$, we obtain

$$\begin{aligned} z_n^2 &= 2\log n - \log\log n - \log(4\pi) + 2x + o(1) \\ &= 2\log n \left(1 - \frac{\log\log n + \log(4\pi) - 2x}{2\log n} + o(1/\log n)\right), \\ z_n &= \sqrt{2\log n} \cdot \left(1 - \frac{\log\log n + \log(4\pi) - 2x}{4\log n} + o(1/\log n)\right). \end{aligned}$$

Therefore, we derive the required sequences:

$$\begin{aligned} a_n &= \sigma\sqrt{2\log n} - \sigma \cdot \frac{\log\log n + \log(4\pi)}{2\sqrt{2\log n}}, \\ b_n &= \frac{\sigma}{\sqrt{2\log n}}. \end{aligned}$$

Problem 7.1

In Section 7.1.2, we have shown that the convergence rate of gradient descent on quadratic objectives is controlled by the following quantity:

$$\rho(\eta) := \max_{i\in[d]} |1-\eta\lambda_i| .$$

Under the optimal choice of η, which is $\eta = \frac{2}{\lambda_{\min}+\lambda_{\max}}$, we have

$$\rho^* = 1 - \frac{2}{1+\kappa}.$$

(a) Suppose you really hate oscillatory behavior and really want to make the convergence entirely stable, which means that you want to ensure $1-\eta\lambda_i \in [0,1)$ for all $i \in [d]$. Show that under the optimal choice of η in this regime, we have $\rho(\eta) = 1 - \frac{1}{\kappa}$.

Note: This means if you insist on having a stable convergence, the number of steps to reach $L(\boldsymbol{\theta}_t) \le L_{\min} + \epsilon$ will be increased from $T(\epsilon) \sim \frac{\kappa}{4}\log\frac{1}{\epsilon}$ (Equation (7.68)) to $T(\epsilon) \sim \frac{\kappa}{2}\log\frac{1}{\epsilon}$.

(b) In practice, we cannot compute the optimal learning rate exactly, since computing the maximum and minimum eigenvalues can be even slower than solving the optimization problem itself. A common practice is to use grid search: start with a base learning rate η_0, and then try gradient descent with learning rate $\eta_k = \alpha^k\eta_0$ for many $k \in \mathbb{Z}$, where $\alpha > 1$ is a constant. Suppose that we are unlucky with the choice of η_0, but we search very thoroughly over k. Then the convergence rate is controlled by

$$\rho_\alpha^* := \sup_{\eta_0>0}\inf_{k\in\mathbb{Z}} \rho\left(\alpha^k\eta_0\right).$$

Express ρ_α^* in terms of α and κ. You can assume that κ is so large that $\alpha > 1 + \frac{1}{\kappa}$.

Note: Typical choices of α in deep learning are $\sqrt{2}$, 2, $\sqrt{10}$, 10, depending on your compute budget.

Answer:

For part (a), to ensure $1-\eta\lambda_i \ge 0$ for all $i \in [d]$, we must have $\eta \le \frac{1}{\lambda_{\max}}$. Under this constraint, the optimal learning rate is $\eta^* = \frac{1}{\lambda_{\max}}$, and the convergence factor satisfies

$$\rho(\eta^*) = 1 - \eta^*\lambda_{\min} = 1 - \frac{\lambda_{\min}}{\lambda_{\max}} = 1 - \frac{1}{\kappa}.$$

For part (b), let $u = \eta_0\lambda_{\max}$. We then have

$$\rho_\alpha^* = \sup_{\eta_0>0}\inf_{k\in\mathbb{Z}} \rho\left(\alpha^k\eta_0\right) = \sup_{\eta_0>0}\inf_{k\in\mathbb{Z}} \max\left(1-\frac{\alpha^k u}{\kappa}, \alpha^k u - 1\right).$$

To maximize the minimum value, the optimal balance condition requires $1 - \frac{u}{\kappa} = \alpha u - 1$ with $u \le u^* = \frac{2\kappa}{\kappa+1} \le \alpha u$. This implies

$$u = \frac{2\kappa}{\alpha\kappa+1}, \quad \rho_\alpha^* = \frac{\alpha\kappa-1}{\alpha\kappa+1}.$$

Since $\alpha > 1 + \frac{1}{\kappa}$, the above bound is achievable.

Problem 7.2

Let $\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top \boldsymbol{H}\boldsymbol{\theta}$ be a quadratic objective, where $\boldsymbol{H} \in \mathbb{R}^{d\times d}$ is symmetric. Assume that $\boldsymbol{H}$ has at least one negative eigenvalue. Show that if we run gradient descent with any learning rate $0 < \eta < \frac{2}{\lambda_{\max}(\boldsymbol{H})}$ and random initialization $\boldsymbol{\theta}_0 \sim N(\mathbf{0}, \boldsymbol{I})$, then for any $M > 0$, with probability 1 we have $\mathcal{L}(\boldsymbol{\theta}_t) < -M$ for some $t \geq 0$.

Note: In general, we say that $\boldsymbol{\theta}$ is a saddle point if $\nabla\mathcal{L}(\boldsymbol{\theta}) = \mathbf{0}$ but $\boldsymbol{\theta}$ is not a local minimizer of L. It can be shown that if $\nabla^2\mathcal{L}(\boldsymbol{\theta})$ has at least one negative eigenvalue at a saddle point $\boldsymbol{\theta}$, then it happens with probability 1 that gradient descent with random initialization will not converge to this saddle point. Check the stable manifold theorem if you are interested.

Answer:

Under this condition, we have $\nabla\mathcal{L}(\boldsymbol{\theta}) = \boldsymbol{H}\boldsymbol{\theta}$; thus the gradient descent update is

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\nabla\mathcal{L}(\boldsymbol{\theta}_t) = (\boldsymbol{I} - \eta\boldsymbol{H})\boldsymbol{\theta}_t.$$

We expand $\boldsymbol{\theta}_t$ in the eigenbasis of $\boldsymbol{H}$. Let $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_d\}$ be an orthonormal eigenbasis of $\boldsymbol{H}$, with eigenvalue λ_i associated to $\boldsymbol{v}_i$. Let $\alpha_{t,1}, \ldots, \alpha_{t,d}$ be the coefficients such that

$$\boldsymbol{\theta}_t = \sum_{i=1}^{d} \alpha_{t,i}\boldsymbol{v}_i.$$

Then we have

$$\alpha_{t+1,i} = (1 - \eta\lambda_i)\alpha_{t,i} = (1 - \eta\lambda_i)^t\alpha_{0,i},$$
$$\mathcal{L}(\boldsymbol{\theta}_t) = \frac{1}{2}\sum_{i=1}^{d} \lambda_i\alpha_{0,i}^2(1 - \eta\lambda_i)^{2t}.$$

By assumption there exists some index $1 \leq i^* \leq d$ satisfying $\lambda_{i^*} < 0$, which implies $1-\eta\lambda_{i^*} > 1$. For random initialization $\boldsymbol{\theta}_0 \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$, the coefficient $\alpha_{0,i^*} \sim \mathcal{N}(0, 1)$ is nonzero with probability one. As a result, the term $\lambda_{i^*}(1 - \eta\lambda_{i^*})^{2t}$ tends to $-\infty$ as $t \to \infty$. For other directions $i \neq i^*$ with $\lambda_i \geq 0$, the bound $\eta < \frac{2}{\lambda_{\max}}$ yields $|1 - \eta\lambda_i| \leq 1$, so their corresponding terms stay uniformly bounded.

Therefore, for any $M > 0$, there exists some t such that $\mathcal{L}(\boldsymbol{\theta}_t) < -M$ with probability one.